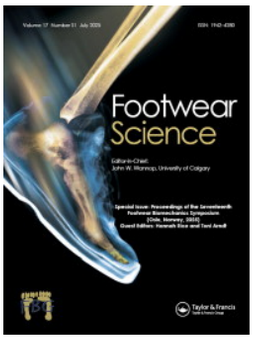

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ABSTRACT

Biomechanical impact of soccer shoe outsole design on knee joint during cutting movements

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KEYWORDS Biomechaincs; studs; knee moment; ACL; soccer shoes

Introduction

The ability to quickly change direction is crucial for scoring in soccer. Sidestep-cutting, a key technique for rapid direction changes, is commonly used in the sport. Research shows that players perform thousands of sidestep-cutting movements in a match, and 6%-9% of ACL injuries occur during these actions (Patel et al., 2023).

Soccer shoes are essential equipment for players, serving as the primary contact between the player and the field. The design of the shoes is important for both performance and injury prevention (Palucci Vieira et al., 2024).

Studs play a key role in traction. The number and distribution of studs help to spread pressure, reduce peak forces, and improve stability and comfort (O'Connor & James, 2013). However, due to differences in foot shape, the ideal stud placement is still unclear. To reduce injury risk, some studies suggest increasing the number of studs and adding protective materials in the forefoot area.

Purpose of the study

The study aims to investigate the biomechanical effects of different stud numbers and distributions on soccer players' ankle joints during sidestep cutting.

Methods

Seventeen healthy male soccer players participated in this study. Two shoe conditions were evaluated in a random order: (1) 8 studs in the forefoot and (2) 9 studs in the forefoot. The shoes had no significant differences in stud shape, upper material, or weight, but varied in stud number and distribution. Participants were instructed to perform a 45° dribbling sidestep cutting motion.

Kinematic data were collected using 12 Vicon MX cameras (Oxford, UK) at 250 Hz, and kinetic data were obtained from a Kistler force plate (Alton, UK) at 1000 Hz. Reflective markers were placed on the lower limbs of the participants. Rigid marker clusters with four non-orthogonal markers were fixed on the thigh and calf. Kinematic and kinetic curves were calculated using a six-degree-of-freedom model in Visual3D software (C-Motion, Inc., Germantown, USA). Both kinematic and kinetic data were filtered using a fourth-order Butterworth filter with a cut-off frequency of 15 Hz. External joint moments were normalised to body weight.

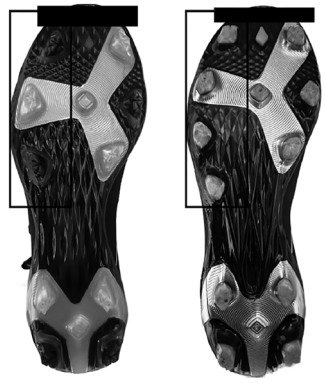

Figure 1. Experimental footwear (left: 8 studs in the forefoot; right: 9 studs in the forefoot).

Table 1. Biomechanical variables in the different footwear conditions. Bold values represent a significant difference ($p < 0.05$).

	ATM1	ATM2	*p*
P_{VGRF} (KG·BW^{-1})	2.64 (0.25)	2.62 (0.39)	0.524
DCT	0.27 (0.03)	0.34 (0.02)	**<0.05**
K_{VAL} Moment (N·m·BW^{-1})	0.83 (0.32)	0.98 (0.27)	**<0.05**
K_{VAR} Moment (N·m·BW^{-1})	0.71 (0.21)	0.94 (0.26)	0.161
K_{IR} Moment (N·m·BW^{-1})	0.79 (0.27)	1.16 (0.56)	**<0.05**
K_{ER} Moment (N·m·BW^{-1})	0.61 (0.33)	0.94 (0.52)	0.087
K_{FLEX} Moment (N·m·BW^{-1})	3.53 (0.53)	3.15 (1.13)	0.209
K_{EXTE} Moment (N·m·BW^{-1})	2.26 (1.80)	2.06 (1.15)	0.177

Note: Statistical significance was set to $p < 0.05$. P_{VGRF}: Peak Vertical GRF; K_{VAL} Moment: knee valgus moment; K_{VAR} Moment: knee varus moment; K_{IR} Moment: knee internal rotation moment; K_{ER} Moment: knee external rotation moment; K_{FLEX} Moment: knee flexion moment; K_{EXTE} Moment: knee extension moment.

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A paired t-test (alpha = 0.05) was used to assess the differences between two conditions using SPSS Statistics (Version 27, IBM, Armonk, USA) (Figure 1).

Results

We observed significant differences in the dynamic traction coefficient (DCT) between ATM1 and ATM2, with values of 0.27 and 0.34, respectively ($p < 0.05$). Similarly, there was a notable difference in knee valgus moment ($p < 0.05$), with ATM2 showing a higher value of 0.98 N·m·BW^{-1}, exceeding ATM1 by 0.15 N·m·BW^{-1}. Furthermore, knee internal rotation moments also differed significantly ($p < 0.05$), with ATM2 registering 1.16 N·m·BW^{-1}, 0.37 N·m·BW^{-1} higher than ATM1 (Table 1).

Discussion and conclusion

Our study provides an initial investigation into the impact of soccer shoe outsole design on knee biomechanics during sidestep cutting. The results suggest that increasing the number of studs in the forefoot can enhance traction, improving stability during sidestep movements. However, this may also increase the risk of knee injuries, particularly ACL injuries.

Although no significant differences were found between the two shoe types in some aspects, ATM2 shoes had a more pronounced effect on knee valgus and varus angles, as well as internal and external rotation angles. Excessive valgus and internal rotation may increase knee joint pressure and rotational stress, potentially raising the risk of knee injuries such as ACL tears (Patel et al., 2023).

Athletes should consider their specific needs and use scenarios when choosing footwear, being mindful of the potential increased joint load associated with higher traction.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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